



7. Arrays

Store similar data together, access it instantly.

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Before we explore more advanced data structures, we should start with one of the most fundamental ones: the **Array**.

An array is one of the simplest and most widely used data structures in computer science. In fact, if you have already worked with Python lists, many array operations will feel very familiar. Arrays provide an efficient way to store and organize collections of related data, making them the foundation of many algorithms and data processing tasks.

Whenever we need to store multiple values of the same type—such as exam scores, temperatures, employee IDs, or pixel values in an image—arrays are often the first structure we consider. Their simplicity, efficiency, and direct access to elements make them essential in programming, data science, machine learning, and software development.

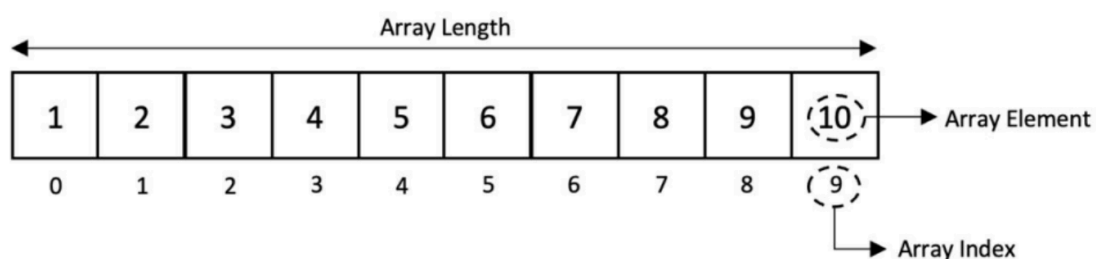
In this section, we will explore what arrays are, how they are stored in memory, how they differ from Python lists, and how we can implement them using NumPy.

1. What is an Array?

Imagine we want to store the marks of 100 students. Creating 100 separate variables would be inefficient and difficult to manage. Instead, we can place all of the values into a single structure called an **array**.

An array is a **linear data structure** that stores multiple values of the same type in consecutive memory locations. Each element occupies a specific position inside the array, allowing us to access it directly using its index.

An array consists of three important components:



- **Array Length (or Size):** The length of an array represents the total number of elements stored inside it. For example, if an array contains 10 numbers, its length is 10.
- **Array Element:** An element is an individual value stored within the array. For example, in an array of student scores: `[85, 92, 78, 90, 88]`. Each number is an array element. Since arrays are designed for efficiency, all elements are typically required to have the same data type, such as all integers, all floating-point values, or all strings.
- **Array Index:** The index identifies the position of an element inside the array. Most programming languages, including Python, use **zero-based indexing**,

meaning the first element is stored at index `0`.

For example: `scores = [85, 92, 78, 90, 88]`

Index	Value
0	85
1	92
2	78
3	90
4	88

To access the first value, we use: `scores[0]`. Which returns: `85`

One of the most important characteristics of an array is that its elements are stored in **contiguous memory locations**.

Think of an array as a row of lockers placed side by side:

```
+-----+-----+-----+-----+-----+
| 85 | 92 | 78 | 90 | 88 |
+-----+-----+-----+-----+-----+
  0     1     2     3     4
```

Because the elements are stored sequentially in memory, the computer can calculate the exact location of any element directly from its index. This allows arrays to provide extremely fast access to data, regardless of the element's position. This property makes arrays highly efficient for: Numerical computations; Scientific simulations; Image processing; Machine Learning; Data analysis

Implementing Arrays using NumPy

In Python, arrays can be implemented using either the built-in **array module** or the **NumPy** package.

Among these options, NumPy is by far the most popular because it provides:

- High-performance numerical operations
- Multi-dimensional arrays
- Vectorized computations
- Mathematical and statistical functions

- Integration with Data Science and Machine Learning libraries

For example:

```
import numpy as np

numbers = np.array([1, 2, 3, 4, 5])
print(numbers)
```

[1 2 3 4 5]

NumPy arrays are optimized for performance and can process large amounts of data much more efficiently than standard Python lists.

Arrays using Python Lists

Although Python does not have a built-in array syntax like some other programming languages, we can often use Python lists to achieve similar behavior.

For example:

```
numbers = [1, 2, 3, 4, 5]
```

This looks similar to an array, but there are several important differences.

Array	List
Elements should have the same data type	Elements can have different data types
Usually fixed in size after creation	Dynamic size that can grow or shrink
More memory efficient	Uses more memory
Faster for numerical operations	Generally slower for numerical processing
Fewer built-in methods	Many built-in methods available

For example, a Python list can contain mixed data types:

```
mixed = [10, "Hello", 3.14, True]
```

While this flexibility is useful, it also introduces additional memory overhead. Arrays sacrifice some flexibility in exchange for better performance and more

predictable memory usage.

As a result, arrays are commonly used when we need efficient storage and fast computation, while lists are preferred when flexibility is more important.

2. Types of Arrays

Arrays can be classified in different ways depending on how they are organized. The two most common classifications are based on **size** and **dimensions**.

Arrays Based on Size

When we talk about the size of an array, we are referring to whether the number of elements can change after the array has been created.

- **Fixed-Size Arrays:** A fixed-size array has a size that is determined when the array is created and cannot be changed afterward. Since memory is allocated in advance, we must know how many elements we intend to store before creating the array.

For example, if we create an array with space for 100 elements:

```
scores = [0] * 100
```

The array reserves space for 100 values.

This approach can create two problems: If we allocate more space than necessary, memory is wasted. If we allocate less space than required, there may not be enough room to store all data.

Because of these limitations, fixed-size arrays are commonly used when the amount of data is known beforehand.

- **Dynamic-Size Arrays:** A dynamic-size array can grow or shrink during program execution. Instead of allocating a fixed amount of memory, the structure automatically adjusts its size when elements are added or removed.

Python lists are a good example of dynamic arrays:

```
numbers = [1, 2, 3]
```

```
numbers.append(4)
```

```
numbers.append(5)
```

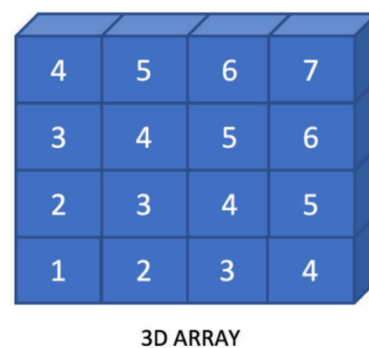
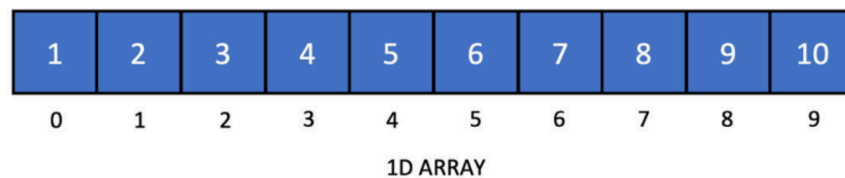
```
print(numbers)
```

[1, 2, 3, 4, 5]

Dynamic arrays provide greater flexibility because programmers do not need to know the final size of the collection in advance.

Arrays Based on Dimensions

Arrays can also be classified according to the number of dimensions they possess. A dimension can be thought of as a direction in which data is organized.



- **One-Dimensional Array (1D Array):** A one-dimensional array is the simplest form of an array. It consists of a single sequence of elements arranged in one direction.

```
numbers = [10, 20, 30, 40, 50]
```

Visualization:

```
+---+---+---+---+---+
| 10 | 20 | 30 | 40 | 50 |
+---+---+---+---+---+
```

Most introductory array operations are performed on one-dimensional arrays.

- Multi-Dimensional Arrays: A multi-dimensional array contains more than one dimension. These arrays are useful when representing structured data such as tables, spreadsheets, images, scientific measurements, or simulation data. Common examples include 2D arrays, 3D arrays, and higher-dimensional arrays.
 - Two-Dimensional Array (2D Array): A two-dimensional array can be viewed as an array of arrays. It is organized into rows and columns, similar to a table or matrix. Example:

```
matrix = [
    [1, 2, 3],
    [4, 5, 6],
    [7, 8, 9]
]
```

```
1 2 3
4 5 6
7 8 9
```

2D arrays are widely used in: Matrices; Spreadsheets; Image processing; Board games; Scientific computations

- Three-Dimensional Array (3D Array): A three-dimensional array can be viewed as a collection of two-dimensional arrays stacked together. Example:

```
cube = [
    [
        [1, 2],
        [3, 4]
    ],
    [
        [5, 6],
        [7, 8]
    ]
]
```

Layer 1

1 2

3 4

Layer 2

5 6

7 8

3D arrays are commonly used for: Medical imaging; Computer graphics; Video processing; Scientific simulations

- Higher-dimensional arrays (4D, 5D, and beyond) are also possible and frequently appear in machine learning, artificial intelligence, physics simulations, and big data applications.

3. Applications of Arrays

Arrays are among the most widely used data structures in computer science because they provide fast access to data and serve as the foundation for many algorithms and systems. Let us explore some common applications.

- **Storing and Accessing Data:** One of the most basic uses of arrays is storing collections of related information. For example, Student grades; Employee salaries; Daily temperatures; Product inventories

```
temperatures = [30, 31, 29, 32, 28]
```

Arrays allow us to access any element quickly using its index.

- **Sorting Algorithms:** Many sorting algorithms use arrays as their primary data structure. Examples include Bubble Sort; Selection Sort; Insertion Sort; Merge Sort; Quick Sort. These algorithms rearrange array elements into ascending or descending order.

```
numbers = [5, 2, 8, 1, 9]
```

After sorting: [1, 2, 5, 8, 9]

- **Searching Algorithms:** Arrays are frequently used with searching techniques. Common searching algorithms include Linear Search and Binary Search. For example, finding a student ID within a list of thousands of records.
- **Building Other Data Structures:** Many advanced data structures are built using arrays as their underlying storage mechanism. Examples include Matrices; Stacks; Queues; Heaps; Hash Tables. Arrays also play an important role in dynamic programming, where intermediate results are stored to improve performance.
- **Signal Processing:** Signals such as sound waves, radio waves, and sensor measurements are often represented as arrays of sampled values. Applications include: Speech recognition; Radar systems; Audio analysis; Communication systems
- **Multimedia Applications:** Images, videos, and audio files are fundamentally stored as arrays. For example, an image can be represented as a matrix of pixels.

$$\text{Pixel Array : } \begin{bmatrix} [RGB] & [RGB] & [RGB] \\ [RGB] & [RGB] & [RGB] \\ [RGB] & [RGB] & [RGB] \end{bmatrix}$$

Applications include: Image editing; Video processing; Computer vision; Animation

- **Data Mining and Big Data:** Large datasets are often stored and manipulated using arrays. Arrays enable fast data retrieval, efficient analysis, statistical processing, and Machine Learning computation.
- **Robotics:** Robotics systems frequently use arrays to represent Sensor readings; Object positions; Movement coordinates; Robot states. These arrays help robots understand and interact with their environment.
- **Real-Time Monitoring and Control Systems:** Many industrial systems continuously collect data from sensors. Examples include manufacturing systems; aerospace systems; smart factories; autonomous vehicles. Arrays allow sensor values to be stored and processed efficiently in real time.
- **Financial Analysis:** Financial applications use arrays to store and analyze stock prices, exchange rates, market indicators, and trading data. These arrays support forecasting, visualization, and automated trading systems.
- **Scientific Computing:** Scientific research relies heavily on arrays to represent numerical data. Applications include physics simulations, weather forecasting, engineering analysis, and biological research. Large-scale scientific computations often involve multi-dimensional arrays containing millions of values.

Advantages of Arrays

Arrays remain one of the most popular data structures because of their simplicity and efficiency.

- **Fast Random Access:** Arrays provide direct access to any element using its index. This operation has a time complexity of $O(1)$, meaning the access time remains constant regardless of the array size.
- **Efficient Memory Layout:** Because elements are stored in contiguous memory locations, arrays allow fast retrieval and efficient memory usage.

- **Support for Various Data Types:** Arrays can store Integers; Floating-point numbers; Characters; Strings; Objects, provided that all elements share the same type in traditional arrays.
- **Simple to Learn and Implement:** Arrays are easy to understand and are usually the first data structure introduced in programming courses.
- **Foundation for Many Algorithms:** Numerous algorithms and data structures depend on arrays for their implementation. Learning arrays provides the foundation for understanding more advanced topics.
- **Hardware Friendly:** Because arrays store data sequentially in memory, they work efficiently with modern computer hardware and processor caches.

Disadvantages of Arrays

Despite their advantages, arrays also have several limitations.

- **Fixed Size Limitation:** Traditional arrays cannot change size after creation. If additional space is needed, a new array must be created and existing elements copied into it.
- **Large Arrays Require Large Memory Blocks:** Allocating a very large array may fail if sufficient contiguous memory is unavailable. This can be problematic in memory-constrained environments.
- **Slow Insertions and Deletions:** When inserting or removing elements in the beginning or the middle of an array, many elements may need to be shifted. For example: The array is `[10][20][30][40]`, insert 25 at the beginning might require shifting every element. The shifting operation can be expensive for large arrays.
- **Potential Memory Waste:** If an array is allocated with more space than necessary, unused memory remains reserved but unused.
- **Limited Flexibility:** Since all elements generally share the same data type and the size is often fixed, arrays are less flexible than structures such as linked lists, trees, and dictionaries.
- **Not Ideal for Complex Relationships:** Arrays are excellent for storing sequential data, but they are not always the best choice when representing complex relationships between objects. For such scenarios, more advanced data structures are often preferred.

Summary

In this section, we explored one of the most fundamental data structures in computer science: the **Array**. We learned that arrays store multiple elements of the same type in contiguous memory locations, allowing for fast and efficient access through indexing. We also examined the key components of an array, including its size, elements, and indexes.

We then classified arrays based on **size** and **dimensions**, distinguishing between fixed-size and dynamic-size arrays, as well as one-dimensional and multi-dimensional arrays. These different forms allow arrays to represent everything from simple lists of values to complex structures such as matrices, images, and scientific datasets.

Beyond their structure, we saw how arrays are used throughout computing, from sorting and searching algorithms to robotics, multimedia processing, financial analysis, scientific computing, and machine learning. Finally, we discussed the major advantages of arrays, such as fast random access and memory efficiency, along with their limitations, including fixed sizing and costly insertions and deletions.

Now that we understand what arrays are, why they are important, and where they are used, we are ready to move on to the practical side of working with them: **performing operations on arrays**. In the next section, we will learn how to create arrays, access elements, update values, insert data, delete elements, and perform other common array operations.

References

https://drive.google.com/file/d/18wXJmAyHVrloU2XspveSV1hGezBzSWEC/view?usp=drive_link

<https://www.geeksforgeeks.org/dsa/array-data-structure-guide/>

Next Sections:

 [7.1. Operations on Arrays](#)

 [7.2. Searching in Arrays](#)

